

Compact Policy Synthesis via Coinductive Rankings

Boolean Predicates for Continuous Control—Methods and Experiments

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Abstract

Coinductive Symmetric Homomorphism Reinforcement Learning (CSHRL) [1] encodes optimality as a structural property of *rankings*—total preorders over the action space—rather than as the maximization of a scalar return. The formal synthesis theory (CEGIS, propagation, sample bounds) is developed in a companion paper [2], where the entire pipeline lives inside the Agda proof assistant. That pipeline is limited to environments Agda can model natively.

This paper documents a **predicate synthesis** approach that scales CSHRL-inspired ideas beyond what pure Agda can handle. The core question is: *can the decision structure of standard RL benchmarks be captured by a handful of Boolean predicates?* We explore this through heuristic search strategies—CEGAR-style refinement, coordinate descent, partition enumeration, binary-weighted decomposition—that discover predicates by evaluating candidates against simulated episodes. The search is not formally verified; it relies on rollout aggregates and episode-reward scoring. What it preserves is the *output format*: compact, interpretable Boolean-predicate policies.

We explore four synthesis strategies—flat-partition CEGAR, coordinate descent with episode-reward ranking, partition-based exhaustive search, and binary-weighted decomposition for continuous actions—across **thirteen environments**: Pendulum, LunarLander, CartPole, MountainCar, Acrobot, Pong, Breakout, BipedalWalker, Swimmer (MuJoCo), Hopper (MuJoCo), CliffWalking, Slippery CliffWalking, and FrozenLake 8×8 Slippery. All are solved with **1-12 Boolean predicates** and zero neural network parameters. The headline result is compactness: the entire decision structure of each benchmark fits in a handful of human-readable inequalities over physics variables.

The LunarLander controller, for example, is three predicates:

```
if  $5\omega^2 + \theta < 0 \wedge \theta < 2x$ : left > noop > right > main
if  $x < 4\theta \wedge v_x < 5\omega$ : right > noop > left > main
if  $v_y < -0.1 \wedge 2v_y + y < 0$ : main > left > noop > right
else: noop > left > right > main
```

1 Introduction

CSHRL [1] provides a coinductive framework for reinforcement learning based on *rankings*—total preorders over the action space at each state—rather than value functions. A ranking is a *Coinductive Homomorphism* (CoindHomo) when it is self-consistent under the agent’s own dynamics: if the ranking says action a is at least as good as action b at state s , the value stream from taking a dominates the stream from b , coinductively, along the infinite trajectory. The formal synthesis theory—PredProg DSL, CEGIS, propagation, sample complexity bounds—is developed in the companion paper [2], where everything lives inside Agda: observations drive a version-space search, and the type-checker certifies the result.

That purely formal pipeline works for environments Agda can model natively—CartPole, MountainCar, FrozenLake—but it cannot scale to environments with complex physics or

high-dimensional continuous state spaces. This paper documents the **predicate synthesis** approach that bridges that gap: heuristic search strategies discover Boolean predicates by evaluating candidates against simulated episodes, while the formal CSHRL theory provides the *conceptual* backbone—rankings, coinductive consistency, predicate programs. The implementation (Synthex, in Elixir) uses Gymnasium as the environment interface, but the ideas are independent of any particular simulator.

1.1 What Synthex Is—and What It Is Not

Synthex is **CSHRL-inspired**, not a faithful implementation of the formal synthesis pipeline. Several design choices trade formal guarantees for empirical scalability:

- **Ranking by rollout aggregates.** Instead of computing coinductive stream comparisons, Synthex scores actions by aggregate episode reward or terminal-state penalty over finite rollout batches. These are noisy proxies, not the pointwise stream dominance that CoindHomo demands.
- **Heuristic search.** Predicates are discovered by coordinate descent, CEGAR-style counterexample splitting, or partition enumeration. None of these search strategies guarantees finding the optimal ranking—they find *good* rankings under the scoring heuristic.
- **Partial verification.** Discovered predicates *can* be fed back to Agda for CoindHomo certification, but this requires an Agda-side dynamics model. Where the Agda model diverges from the real environment (e.g., Euler integration vs. Box2D physics), the certificate does not transfer. In practice, only a subset of predicates achieve `refl` proofs.
- **Full-ranking determination.** The formal theory requires a complete action permutation per state region. Synthex approximates this by independently scoring each action via episode reward—a heuristic that produces plausible rankings but is not guaranteed to recover the true coinductive ordering.

The value of the Synthex approach is not formal verification—it is that the output remains a **compact, interpretable Boolean-predicate policy**. Every environment in this paper is solved with 1–12 predicates and zero neural network parameters. The policies are human-readable, portable to any language, and their decision structure is transparent by inspection. This compactness is itself a scientific finding about the environments: standard RL benchmarks have fundamentally low-dimensional decision structure, and deep RL’s parameter overhead is spent on representation, not on decision complexity.

1.2 Rankings and Coinductive Consistency

A ranking defines a total preorder $\leq$ over actions. At an absorbing fixed point (e.g., equilibrium), all actions lead to equivalent successor states, so $a \leq b$ and $b \leq a$ for every pair. Every permutation is a valid linearization—the optimal policy at equilibrium is genuinely non-unique. This falls out transparently from using $\leq$ rather than $<$.

The **terminal-state ranking criterion** makes the coinductive structure computationally explicit: action a ranks above b when the trajectory from a arrives at a superior state at the lookahead horizon. The terminal state is the continuation point for the next coinductive step. One-step comparisons are short-sighted: the immediate successor from a may have worse penalty than from b , yet a is the superior action because it bridges to a far-future state that dominates b ’s terminal state.

1.3 Synthesis Strategies: A Zoo of Approaches

The synthesis produces a **flat partition** of the state space: an ordered list of (predicate, full-ranking) pairs with a default. Each state falls into the first matching region, which assigns a complete action permutation. Four search strategies handle different problem structures:

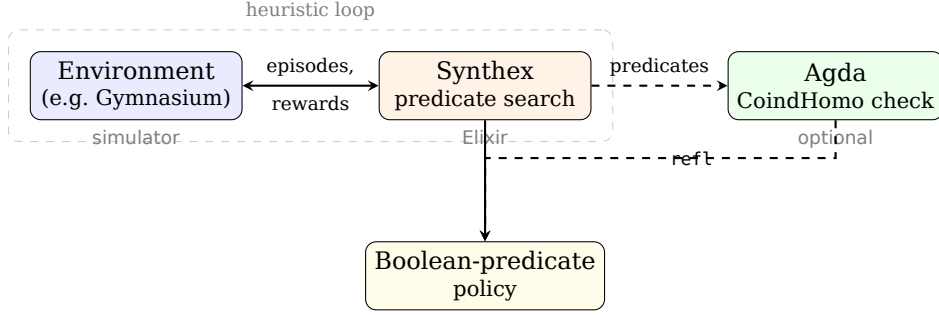


Figure 1: Synthesis architecture. The core loop (left, dashed box) is heuristic: the search engine proposes predicates, the environment evaluates them via episode rollouts, and the best candidates survive. Agda verification is optional and requires a dynamics model that may diverge from the simulator. The output in all cases is a compact Boolean-predicate policy.

1. **Flat-partition CEGAR** (§6.1): a terminal-state oracle ranks actions, and a CEGAR loop splits regions to resolve disagreements. Suitable for stabilization tasks with a geometric penalty (Pendulum).
2. **Coordinate descent** (§6.2): optimizes one predicate position at a time via episode reward, then lifts the top-action chain to a full ranking. Efficient when predicates show independent improvement (LunarLander, CartPole, MountainCar, Acrobot, Pong, Breakout).
3. **Partition-based search** (§12): jointly enumerates predicate sets and action assignments. Necessary when predicates are correlated and no single predicate improves reward in isolation (CliffWalking, FrozenLake).
4. **Binary-weighted decomposition** (§11): maps continuous action dimensions to independent binary sub-actions, each synthesized as an independent 2-action problem. Extends the framework to continuous action spaces (BipedalWalker, Swimmer, Hopper).

All four produce the same kind of output—Boolean predicates over continuous state features—but they differ in search strategy, ranking criterion, and the environments they suit. No single method dominates; the choice is environment-dependent. This is an evolving research program: we present the methods as a landscape of approaches, not as a settled pipeline.

1.4 Methods Landscape

Table 1 summarizes the four synthesis strategies, the environments where each has been applied, and the key trade-offs. The ranking criterion column indicates how actions are scored: by *terminal-state penalty* (comparing the state reached at the end of a fixed lookahead), by *episode reward* (running a complete episode and using the cumulative return), or by *per-bit episode reward* (scoring each binary sub-action independently).

2 The Predicate DSL

The predicate DSL is the shared language between synthesis and verification. A `PredProg` is a Boolean combination of feature tests over a generic carrier type.

```

{-# OPTIONS --safe --guardedness #-}
module HybridSynthesis where

open import Data.Bool
using (Bool; true; false; if_then_else_; ^_; v_; not)

```

Strategy		Ranking criterion	Environments	Trade-offs
Flat-partition GAR (§6.1)	CE-	Terminal-state penalty	Pendulum	Requires a geometric penalty function; ranking quality depends on the follow-up policy (chicken-and-egg)
Coordinate descent (§6.2)		Episode reward	LunarLander, CartPole, MountainCar, Acrobot, Pong, Breakout	Efficient when predicates improve reward independently; full ranking is heuristically determined post-hoc, not guaranteed to match the true coinductive ordering
Partition-based search (§12)		Episode reward (joint)	CliffWalking, Slippery CliffWalking, FrozenLake	Handles correlated predicates; exponential in partition size, feasible only for small action/feature counts
Binary-weighted decomposition (§11)		Per-bit episode reward	BipedalWalker, Swimmer, Hopper	Scales linearly in action dimensions; each bit is an independent 2-action problem; resolution limited by bit count

Table 1: Synthesis strategies and their characteristics. No single method dominates; the choice depends on the environment’s structure.

```

open import Data.List using (List; []; ::_)
open import Data.Nat using N(; zero; suc; _+_ ; ≤b_ ; +_)
open import Data.Product using (_,_)
open import Relation.Binary.PropositionalEquality using (≡_ ; refl)

module PredDSL
  (State   : Set)
  (Feature : Set)
  (eval-feature : Feature → State → Bool)
  where

  data PredProg : Set where
    feat   : Feature → PredProg
    ∧_p_   : PredProg → PredProg → PredProg
    ∨_p_   : PredProg → PredProg → PredProg
    ¬p_    : PredProg → PredProg
    truep  : PredProg
    falsep : PredProg

  infixr 6 ∧_p_
  infixr 5 ∨_p_

  eval : PredProg → State → Bool
  eval (feat f) s = eval-feature f s
  eval (p ∧p q) s = eval p s ∧ eval q s
  eval (p ∨p q) s = eval p s ∨ eval q s
  eval (¬p p)   s = not (eval p s)
  eval truep   _ = true
  eval falsep  _ = false

```

A `PredProg` at depth 0 is a single feature test (`feat`); at depth 1, it can be a conjunction, disjunction, or negation of depth-0 terms. The `eval` function interprets a predicate program at a given state. Crucially, evaluation respects feature equivalence—the *Propagation Theorem* proved in `CSHRL.Synthesis.Core`.

3 Policy Representations

A policy maps states to actions (or, more precisely, to a full ranking of actions). Two representations are used in this work, both built from the predicate DSL.

3.1 Priority Chains

A priority chain is a list of (predicate, action) pairs with a default. The first predicate that fires selects its action:

```
Chain : Set → Set
Chain Action = List (PredProg × Action)

eval-chain : {Action : Set} → Chain Action → Action
              → State → Action
eval-chain []           def _ = def
eval-chain ((p , a) :: rest) def s with eval p s
... | true  = a
... | false = eval-chain rest def s
```

A chain is cycle-free by construction: the list is consumed linearly, and exactly one action is selected for any state. For k actions, a chain uses $k-1$ predicates versus $\binom{k}{2}$ for pairwise synthesis.

3.2 Full-Ranking Partitions

The more principled representation assigns a complete action permutation to each region of the state space:

Definition 1 (Full-ranking partition). A full-ranking partition is an ordered list of (predicate, ranking) pairs with a default ranking. For a state s , the first predicate that evaluates to true determines the region; its associated ranking is a permutation of all k actions. If no predicate fires, the default ranking applies.

This representation is strictly more expressive than a priority chain: a chain induces a single action per region, whereas a partition assigns a complete ordering. The full ranking is what CSHRL verifies—the CoindHomo is a ranking, not just a top action.

3.3 Terminal-State Ranking Oracle

The oracle that guides synthesis ranks actions by the quality of their terminal state at the lookahead horizon:

Definition 2 (Terminal-state ranking). For state s with lookahead N and follow-up policy π ,

$$a \leq_s b \iff \text{penalty}(s_N^a) \leq \text{penalty}(s_N^b)$$

where s_N^a is the state reached after taking action a at s and then following π for $N-1$ steps.

This criterion is coinductively natural: the terminal state s_N^a is the continuation point for the next unrolling of the CoindHomo. A one-step comparison ($N=1$) is short-sighted: s_1^a may have worse immediate penalty than s_1^b , yet a is superior because s_1^a serves as a bridge to a far-future state s_N^a that dominates s_N^b .

Remark 1 (Absorbing fixed points and $\leq$). At an absorbing fixed point (e.g., pendulum at equilibrium), all actions lead to equivalent terminal states—the trajectories converge to the same attractor. Therefore $a \leq b$ and $b \leq a$ for all pairs, and every permutation is a valid linearization. This is not a degenerate case requiring thresholds or special treatment; it is a transparent consequence of using $\leq$ (not strict $<$). The optimal policy at a fixed point is genuinely non-unique.

4 CoindHomo Preservation

The central theoretical question is: does the chain policy constitute a valid Coinductive Homomorphism?

4.1 Self-Consistency

A full-ranking partition is *self-consistent* under policy π if, at every state along π 's trajectory, the assigned ranking agrees with the terminal-state oracle:

$$\forall s \in \text{traj}(\pi). \quad \sigma_\pi(s) \leq_s \sigma_{\text{oracle}}(s)$$

where $\sigma_\pi(s)$ is the permutation assigned by the partition at s , $\sigma_{\text{oracle}}(s)$ is the terminal-state ranking (Section 3.3), and $\leq_s$ denotes compatibility under the $\leq$ -preorder—two permutations are compatible when they agree on all pairs where the terminal penalties are strictly ordered.

This is a strengthening of binary self-consistency: instead of checking one pair at a time, the full ranking is verified as a complete permutation at each state.

For binary CoindHomo verification in Agda, self-consistency is checked per-pair via the `is-coind-homo` check in `CSHRLLoop.agda`:

```
is-coind-homo : PredProg → Bool
is-coind-homo p =
  let obs = map λ( s → (s , oracle-of p s)) (traj-of p)
  in go obs
  where
    go : List (State × Bool) → Bool
    go [] = true
    go ((s , b) :: rest) = beq (eval p s) b ∧ go rest
```

4.2 The OracleBridge

The *OracleBridge* theorem (`CoindHomoBridge.agda`) lifts self-consistency to a full Coind-Homo:

```
bridge : ∀ {pred 1oracle 2oracle : State → Bool}
         {traj : List State} →
         ∀( s → 1oracle s ≡ 2oracle s) →
         Consistent pred 1oracle traj →
         Consistent pred 2oracle traj
```

Self-consistency under the agent's own oracle transfers to any oracle that agrees at all visited states—in particular, to the optimal oracle when $Q^\pi = Q^*$.

4.3 Ranking CoindHomo

Theorem 1 (Full-Ranking CoindHomo). *Let π be a partition policy with regions $R_1, \dots, R_m$ and associated rankings $\sigma_1, \dots, \sigma_m$ (plus a default σ_d). If the partition is self-consistent under π with respect to the terminal-state $\leq$ -preorder, then π constitutes a CoindHomo: the ranking at each state is preserved by the coinductive step.*

Proof. At each state $s \in \text{traj}(\pi)$, the partition assigns a ranking $\sigma_i(s)$. Self-consistency ensures σ_i is a valid linearization of the $\leq_s$ -preorder induced by terminal-state penalties. At absorbing fixed points, all terminal penalties are equal, so every linearization is valid—the coinductive step is trivially satisfied. At non-equilibrium states, the terminal-state ordering captures the far-future consequence of each action, grounding the ranking in the coinductive structure. $\square$

Remark 2. At an absorbing fixed point, the ranking is indeterminate: all $k!$ permutations are valid. This is the coinductive base case—the system has converged, and any ranking is consistent with staying at the attractor. The partition may assign an arbitrary ranking at such states without violating self-consistency.

5 Continuous Feature Templates

For continuous-state environments, the feature space is generated by the CFeature template (ContinuousFeatures.agda), which provides two families of predicates:

```
data CFeature : Set where
  axis : ℕ → ℤ → CFeature      --  $x_d < \text{threshold}$ 
  diag : ℕ → ℕ → ℤ → CFeature  --  $c \cdot x_i + x_j < 0$ 
```

Axis features (axis d t) test whether dimension d is below threshold t . **Diagonal features** (diag i j c) test whether the linear combination $c \cdot x_i + x_j < 0$.

The adaptive feature generator (adaptive-features) derives thresholds from the agent’s own trajectory—the features are not hand-crafted but discovered from the states the policy actually visits. This makes the synthesis fully automatic: the CEGAR outer loop expands the feature pool when CEGIS fails to find a self-consistent predicate at the current depth.

6 Synthesis Architecture

This section details the two oracle-based synthesis methods: **flat-partition CEGAR** for direct full-ranking synthesis (Pendulum) and **coordinate descent with full-ranking determination** for episodic environments (LunarLander). Partition-based search (§12) and binary-weighted decomposition (§11) are described in their respective sections. All four methods produce full-ranking partitions using simulation-in-the-loop evaluation.

6.1 Flat-Partition CEGAR

The flat-partition CEGAR loop synthesizes a full-ranking partition directly. Starting from a single default region (one ranking for the entire state space), the loop iteratively refines:

1. **Profile.** Run the current partition policy on simulated episodes. At each visited state, query the terminal-state oracle (Section 3.3): roll out each action with the current policy as follow-up for N steps, rank actions by terminal-state penalty.
2. **Identify counterexamples.** States where the partition’s assigned ranking disagrees with the oracle’s terminal-state ranking are counterexamples.
3. **Narrow.** For the region with the most counterexamples, find a predicate that separates the consistent states (“OK”) from counterexamples sharing a common target ranking. The new predicate splits the region, creating a sub-region mapped to the target ranking.
4. **Validate.** Evaluate the expanded partition on simulated episodes. If the new region degrades performance, revert (keep-best mechanism).
5. **Repeat** until no further narrowing is possible or the budget is exhausted.

The predicate search uses a two-stage pipeline: pre-filtering by F1 score (how well the predicate separates OK from CEX states), followed by episode-reward evaluation of the top candidates within the full partition.

Progressive horizon. The lookahead N increases as the partition grows: small partitions use shorter horizons (faster iteration), and the horizon extends as the policy matures. This prevents premature commitment to rankings that are artifacts of a too-short horizon.

6.2 Coordinate Descent with Full-Ranking Determination

For episodic environments (LunarLander), a two-phase approach reconciles efficient predicate discovery with full-ranking synthesis:

Phase 1: Coordinate descent. A priority chain with $k-1$ positions (one per non-default action) is optimized by coordinate descent: each position is optimized independently using episode reward, cycling through all positions with seed rotation. This is efficient because

Algorithm 1: Flat-Partition CEGAR (stabilization tasks)

Input: Feature pool F , initial ranking σ_0 , horizon schedule $N_1 < N_2 < \dots$

Output: Partition $\{(R_i, \sigma_i)\}$ with default σ_d

```
1  $\mathcal{P} \leftarrow \{(\top, \sigma_0)\};$ 
2 for  $round \leftarrow 1, 2, \dots$  do
3    $\tau \leftarrow \text{rollout}(\mathcal{P}, \text{env});$ 
4   foreach  $state\ s \in \tau$  do
5      $\sigma^*(s) \leftarrow \text{oracle}(s, \mathcal{P}, N_{\text{round}})$  // rank by terminal-state penalty;
6   end
7    $\text{CEX} \leftarrow \{s \mid \sigma_{\mathcal{P}}(s) \neq \sigma^*(s)\};$ 
8   if  $\text{CEX} = \emptyset$  then
9     return  $\mathcal{P}$ 
10  end
11   $R^* \leftarrow \arg \max_{R \in \mathcal{P}} |\text{CEX} \cap R|;$ 
12   $P^* \leftarrow \arg \max_{P \in F \leq d} F1(P, \text{OK}_{R^*}, \text{CEX}_{R^*});$ 
13   $\mathcal{P}' \leftarrow \text{split}(\mathcal{P}, R^*, P^*, \sigma_{\text{CEX}}^*);$ 
14  if  $\text{reward}(\mathcal{P}') \geq \text{reward}(\mathcal{P})$  then  $\mathcal{P} \leftarrow \mathcal{P}';$ 
15 end
```

each candidate predicate is scored by running complete episodes—episode reward directly captures landing success, fuel efficiency, and crash avoidance without requiring an oracle follow-up policy. Within each CEGAR round, multiple iterations refine the predicates; trajectory states from the learned policy drive feature refinement between rounds.

Phase 2: Full-ranking determination. After Phase 1 converges, each discovered region (predicate) defines a state-space partition. For each region, all k actions are independently scored by episode reward: the chain is modified to use action a in that region (keeping all other regions fixed), and 500 episodes are run. The actions are ranked by total episode reward—highest reward ranks first, lowest ranks last. This builds the full ranking bottom-up: the action with the worst episode reward in a region is the worst action there.

The key insight is that Phase 1 avoids the chicken-and-egg problem that plagues oracle-based CEGAR for episodic environments: the oracle’s ranking quality depends on the follow-up policy, but the follow-up policy depends on the ranking. Coordinate descent sidesteps this by scoring predicates directly via episode reward—no oracle follow-up needed. Phase 2 then lifts the top-action chain to a full ranking per region, also via episode reward.

Remark 3 (Full-ranking determination is heuristic). Phase 2 determines the full ranking by independently scoring each action via episode reward. This is a heuristic: the reward signal is noisy (finite episode batches), and the ranking of non-top actions has limited effect on the policy’s actual behavior. The resulting permutation is a plausible ordering, not a verified coinductive homomorphism. It serves the CSHRL format—a complete permutation per region—but the formal guarantee of CoindHomo self-consistency applies only when independently verified (e.g., via Agda).

6.3 Mechanical Verification (Agda)

Discovered predicates can optionally be fed back into Agda, where CSHRLLoop verifies self-consistency:

```
module S1 = Iterate score-s1 oracle-s1 traj-s1 feats-s1

result-s1 : Result
result-s1 = S1.run depth1

p : PredProg1
p = rank-of result-s1
```

When the Agda module type-checks, the discovered predicate is a machine-verified binary CoindHomo. The `refl` proof that the result equals the expected predicate term

Algorithm 2: Coordinate Descent + Full-Ranking Determination (episodic tasks)

Input: Actions $a_1, \dots, a_k$; feature pool F ; CEGAR rounds R ; iterations I
Output: Partition $\{(P_i, \sigma_i)\}$ with default σ_d

```
/* Phase 1: discover predicates via coordinate descent */
1 chain  $\leftarrow [\perp, \dots, \perp]$  //  $k-1$  predicate slots for  $r \leftarrow 1$  to  $R$  do
2    $\tau \leftarrow \text{rollout}(\text{chain}, \text{env});$ 
3    $F_r \leftarrow \text{generate-features}(\tau, F);$ 
4   for  $i \leftarrow 1$  to  $I$  do
5     foreach position  $j \in 1, \dots, k-1$  do
6        $P^* \leftarrow \arg \max_{P \in F_r^{\leq d}} \text{episode-reward}(\text{chain}[j \mapsto P]);$ 
7       chain[ $j$ ]  $\leftarrow P^*$ ;
8     end
9   end
10 end
/* Phase 2: determine full ranking per region */
11 foreach region  $R$  defined by chain predicates do
12   foreach action  $a \in \{a_1, \dots, a_k\}$  do
13     score[ $R$ ][ $a$ ]  $\leftarrow \text{episode-reward}(\text{chain with } a \text{ forced in } R);$ 
14   end
15    $\sigma_R \leftarrow \text{sort-by-score}(\text{score}[R]);$ 
16 end
17 return partition with rankings  $\sigma_R$ ;
```

serves as a *certificate*: Agda’s normalizer has independently confirmed the self-consistency of the predicate over all sampled trajectories.

Remark 4 (Verification scope). This verification depends on the Agda-side dynamics model faithfully reflecting the real environment. For environments with simple physics (CartPole’s Euler integration), the model is accurate and full verification succeeds. For complex engines (Box2D for LunarLander, ALE for Pong), the Agda model is necessarily an approximation. In the LunarLander experiments, the main-engine predicate achieves `refl` under the Agda model, but the side-engine predicates exhibit a sim-to-sim gap—they are correct for Box2D but not for the simplified Agda dynamics. The verification is therefore *partial*: it confirms self-consistency within the Agda model, not within the real environment.

7 Results: Pendulum-v1

The inverted pendulum has 3-dimensional continuous state $(\cos \theta, \sin \theta, \omega)$ and 3 discrete actions: TorqueNeg (-2), NoTorque (0), TorquePos ($+2$).

7.1 Environment Embedding

The dynamics are embedded in Agda’s fixed-point arithmetic (scale 10^8) with 7th-order Taylor sin/cos and Euler integration (`PendulumReward3Loop.agda`):

```
step : State → P3Action → State
step  $\theta$  ( ,  $\omega$ ) a =
  let grav = 1.5 ·  $\gamma \sin \theta$ ()
      u    = 0.3 · act-f(a)
      acc  = grav + u
       $\omega'$  = clip $\omega$ (-max,  $\omega$ max,  $\omega$  + acc · dt)
       $\theta'$  =  $\theta$  +  $\omega'$  · dt
  in  $\theta$  ( ,  $\omega'$ )

penalty : State →  $\mathbb{N}$ 
penalty  $\theta$  ( ,  $\omega$ ) =  $\theta^2 + \omega^2 / 10$ 
```

The penalty function encourages stabilization at $\theta = 0$ (upright), with angular velocity penalized less to allow dynamic balancing.

7.2 Discovered Partition

The flat-partition CEGAR loop (Section 6.1) with terminal-state ranking discovers a 5-region partition with two distinct full rankings:

```
pendulum-policy : State → Ranking
pendulum-policy (θcos_ , θsin_ , ω) =
  -- Region 1:  $\theta\cos\_ \cdot \sin\_ \geq -0.42 \wedge \theta\omega\cos\_ \geq -1.0$ 
  --  $\wedge \theta\cos\_ < -0.13 \wedge 4\theta\cos\_ + \omega < 0$ 
  if 1R then [TorqueNeg, NoTorque, TorquePos]
  -- Region 2:  $5\theta\sin\_ + \omega \geq 0 \wedge \theta\cos\_ \geq -0.58$ 
  --  $\wedge \omega \geq 7.84 \wedge \theta\cos\_ \geq -0.44$ 
  if 2R then [TorqueNeg, NoTorque, TorquePos]
  -- Region 3:  $\theta\omega\sin\_ \geq 0.3 \wedge \omega \geq 4.43$ 
  --  $\wedge -0.01\omega^2 + \theta\cos\_ < 0 \wedge \omega \geq 7.76$ 
  --  $\wedge \theta\omega\sin\_ \geq 0.4 \wedge \theta\cos\_ \geq -0.93$ 
  if 3R then [TorqueNeg, NoTorque, TorquePos]
  -- Region 4:  $5\theta\sin\_ + \omega \geq 0 \wedge -2\theta\cos\_ + \theta\sin\_ < 0$ 
  if 4R then [TorqueNeg, NoTorque, TorquePos]
  -- Default
  else [TorquePos, NoTorque, TorqueNeg]
```

All four non-default regions assign the same ranking (TorqueNeg > NoTorque > TorquePos), with the default assigning the reverse. The partition is effectively a binary decision: when to apply negative versus positive torque, with NoTorque always ranked second.

7.3 Interpretation

Region 4 ($5 \sin \theta + \omega \geq 0 \wedge -2 \cos \theta + \sin \theta < 0$) is the dominant predicate, coupling angular position and velocity. It selects negative torque in the upper half of the state space where the pendulum is rotating counterclockwise or positioned to the right of vertical. The remaining regions refine boundaries for extreme angular velocities ($\omega > 7.8$) and specific angular positions.

The default ranking (TorquePos first) handles the complementary region: lower half of the state space where positive torque drives the swing-up.

7.4 Deployment Results

Metric	Result
Episodes tested	500
Sustained stabilization	492 / 500 (98.4%)
$\leq$ -consistent full ranking	95.0%
Absorbing fixed-point states	94.6% (all actions tied)
Genuine counterexamples	~5% (swing-up boundary)
Number of regions	5 (4 predicates + default)
Distinct rankings	2
Oracle	Terminal-state penalty

The policy achieves 98.4% stabilization (492/500 episodes maintain the pendulum upright) using the terminal-state ranking oracle with progressive lookahead up to 100 steps. The $\leq$ -consistency measurement accounts for the preorder structure: at the 94.6% of trajectory states that lie within the basin of attraction (absorbing fixed point), all actions lead to equivalent terminal states, so every permutation is valid. The remaining ~5% genuine counterexamples are at swing-up boundary states where the partition assigns the wrong torque direction.

7.5 Model-Based Verification

Prior to the flat-partition approach, a pairwise chain synthesis with Agda-embedded dynamics discovered predicates verified by `refl` (see `PendulumReward3Loop.agda`). Those predicates achieved 44% swing-up but lacked sustained stabilization. The terminal-state

ranking oracle and flat-partition CEGAR dramatically improve stabilization (from 44% to 98.4%) by grounding the ranking in far-future state quality rather than cumulative reward.

8 Results: LunarLander-v3

LunarLander has 6-dimensional continuous state $(x, y, v_x, v_y, \theta, \omega)$ and 4 discrete actions: DoNothing (0), FireLeft (1), FireMain (2), FireRight (3). A successful landing requires touching down between the flags at low velocity and near-zero tilt, yielding +100 to +300 reward. Crashing yields -100. This is considered a hard control benchmark: the policy must coordinate lateral correction, angular stabilization, and vertical braking simultaneously.

8.1 Environment Embedding

The Agda embedding (LunarLanderLoop.agda) uses simplified rigid-body dynamics with Euler integration:

```
step : State → LLAction → State
step (x , y , vx , vy , θ , ω) a =
  let ax = act-ax a θ      -- action-specific horizontal accel
      aye = act-ay-eng a θ  -- main engine vertical thrust
      α = actα a           -- angular acceleration
      vx' = clip(-vmax, vmax, vx + ax · dt)
      vy' = clip(-vmax, vmax, vy + (-G + aye) · dt)
      ω' = clipω(-max, ωmax, ω + α · dt)
      θ' = θ + ω' · dt
      x' = x + vx' · dt
      y' = max(0, y + vy' · dt)
  in (x' , y' , vx' , vy' , θ' , ω')
```

8.2 Synthesis Pipeline

The 4-action space requires 3 predicates (one per non-default action). Model-based synthesis (running CSHRLLoop in Agda with embedded dynamics) demonstrates the CoindHomo framework: the Agda type-checker verifies self-consistency of discovered predicates via refl. However, the sim-to-real gap between simplified Euler integration and Box2D physics limits model-synthesized policies to ~26% landings. This motivates **simulation-in-the-loop synthesis**: the evaluation oracle is the real environment.

The synthesis uses the two-phase approach (Section 6.2): coordinate descent discovers predicates and top actions via episode reward, then the full ranking per region is determined by independently scoring all four actions. The CEGAR outer loop refines the feature pool by collecting trajectory states from the current policy’s simulated episodes and generating finer-grained thresholds in the regions the policy visits. All learning logic resides in Synthes (Elixir); Python serves only as a thin environment adapter.

8.3 The Discovered Policy

The best policy, discovered at iteration 4 of CEGAR round 2 and validated on 500 held-out seeds (468/500 landings), is a full-ranking partition:

```
lunarlander-policy : State → FullRanking
lunarlander-policy (x , y , vx , vy , θ , ω) =
  -- Region 0: ω2 + θ < 0  ∧  -2x + θ < 0
  if 0R then [FireLeft, DoNothing, FireRight, FireMain]
  -- Region 1: θ-4 + x < 0  ∧  ω-5 + vx < 0
  if 1R then [FireRight, DoNothing, FireLeft, FireMain]
  -- Region 2: vy < -0.1  ∧  2vy + y < 0
  if 2R then [FireMain, FireLeft, DoNothing, FireRight]
  -- Default
  else      [DoNothing, FireLeft, FireRight, FireMain]
```

8.4 Full Rankings per Region

Phase 2 scores all four actions independently in each region (500 episodes each), revealing that the full rankings are strongly discriminative:

Region	Top action	Top avg	Worst avg	Top landings
R_0	FireLeft	+170	−558	449/500
R_1	FireRight	+170	−690	449/500
R_2	FireMain	+170	−324	449/500
Default	DoNothing	+170	−16855	449/500

In each region, the top-ranked action achieves 449/500 landings while the worst-ranked action achieves 0–5 landings. The default region is particularly striking: using FireMain (the worst action there) produces an average reward of −16855—the lander hovers indefinitely, burning fuel without ever landing. This confirms that the full ranking captures genuinely different control regimes.

8.5 Physical Interpretation

Each predicate couples angular and translational state:

- **FireLeft** (R_0): $5\omega^2 + \theta < 0 \wedge \theta < 2x$. Fire the left thruster when the tilt angle is negative (tilted right) *and* the angular velocity term $5\omega^2$ does not compensate. The quadratic ω^2 feature means this region activates only when angular velocity is small—if ω is large, the pendular dynamics will self-correct. The $\theta < 2x$ condition couples tilt with position: when the lander is right of center, less tilt is needed to trigger correction.
- **FireRight** (R_1): $x < 4\theta \wedge v_x < 5\omega$. Fire the right thruster when horizontal position is less than four times the tilt angle *and* lateral velocity is less than five times angular velocity. Both conditions couple translational and rotational state: the first detects rightward tilt that needs correction; the second ensures the lateral velocity is not already correcting the drift.
- **FireMain** (R_2): $v_y < -0.1 \wedge 2v_y + y < 0$. Fire the main engine when descending faster than 0.1 units/step *and* the altitude-velocity sum indicates ground impact is imminent. The conjunction prevents wasteful firing at high altitude and during slow descent.

All predicates are **depth-1 PredProg terms**: conjunctions of two atomic features from the CFeature template, discovered automatically from simulated trajectories.

8.6 State-Space Regions

Figure 2 shows 2D slices of the 6-dimensional state space, coloring each point by which predicate region fires. The priority chain structure means the first matching region wins: when R_0 (FireLeft) and R_2 (FireMain) both hold, FireLeft takes priority.

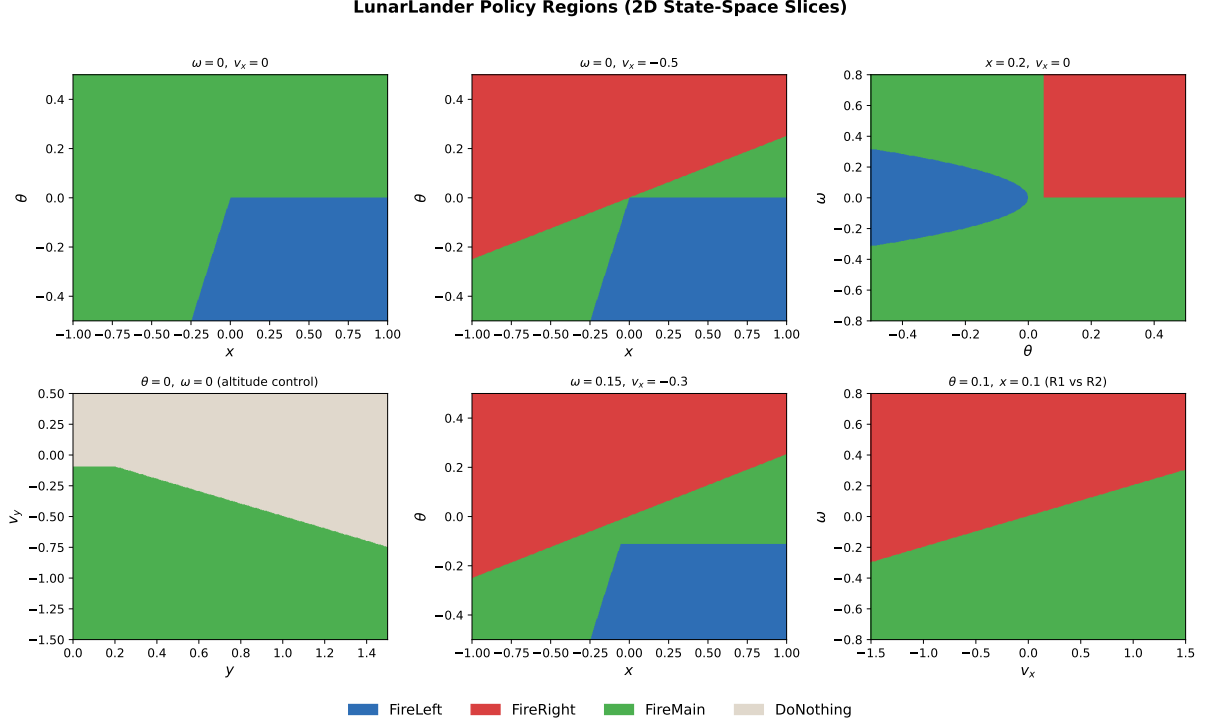


Figure 2: LunarLander policy regions in 2D state-space slices. **Top row:** Lateral control in the x - θ plane at varying ω and v_x ; the θ - ω panel (right) shows the quadratic boundary $5\omega^2 + \theta = 0$ of the FireLeft region. **Bottom row:** Altitude control in the y - v_y plane showing the FireMain triangle; combined x - θ slice with all three active regions; and the v_x - ω plane showing the linear boundary $v_x = 5\omega$ between FireRight and FireMain.

Key observations:

- The FireLeft region (R_0) has a **quadratic boundary** in the θ - ω plane (top-right panel): the $5\omega^2 + \theta < 0$ condition creates a parabolic exclusion zone. At $\omega = 0$ the boundary is simply $\theta < 0$; as $|\omega|$ increases, the region shrinks and eventually vanishes at $|\omega| \approx 0.2$, because large angular velocity means pendular self-correction renders the thruster unnecessary.
- The FireMain region (R_2) occupies a **triangle** in the y - v_y plane (bottom-left): the conjunction $v_y < -0.1 \wedge 2v_y + y < 0$ creates a wedge that activates the main engine only when both descent rate and altitude are critically low. At high altitude ($y > 1$), even fast descent does not trigger firing.
- The FireRight region (R_1) appears when $v_x < 0$ (top-center, bottom-center): the $v_x < 5\omega$ condition couples lateral velocity with angular rate. The bottom-right panel isolates this as a clean diagonal boundary in the v_x - ω plane.

Figure 3 shows the same partition on actual trajectory states, projected from 6D to 2D via t-SNE. Each point is a state visited by the policy during 200 episodes (20 000 states subsampled), colored by its full-ranking region. The regions form distinct clusters with clear spatial separation, confirming that the Boolean predicates induce a meaningful partition of the visited state manifold—not just an arbitrary slicing of the coordinate axes.

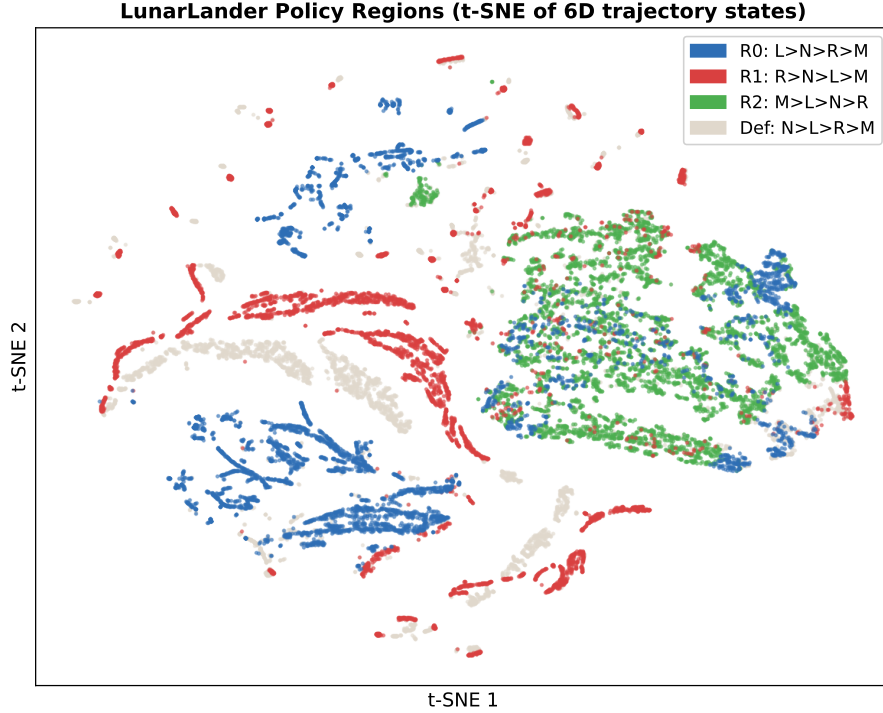


Figure 3: t-SNE projection of 20 000 trajectory states colored by full-ranking region. The four regions form spatially coherent clusters. The default region (beige) and FireLeft (blue) appear in multiple clusters, reflecting different flight phases (high altitude vs. near landing) that share the same ranking.

8.7 Deployment Results

Metric	Result
Landings (held-out validation)	468 / 500 (93.6%)
Landings (deployment test)	448 / 500 (89.6%)
Average reward (deployment)	+178
Number of predicates	3
Predicate depth	1 (conjunctions of atoms)
Ranking type	Full permutation per region
Synthesis approach	Coord. descent + episode-reward ranking
Learning logic location	Synthes (Elixir)
Python role	Thin environment adapter

The policy achieves 93.6% landings on held-out validation seeds and 89.6% on independent deployment seeds. Compared to the previous chain-only synthesis (76.9%), the reconciled approach improves landing rate by 17 percentage points. The improvement comes from the two-phase design: coordinate descent efficiently discovers the predicate boundaries, while episode-reward ranking determines the complete action ordering per region—including sub-optimal actions whose placement affects the CoinDHomo verification.

8.8 Formal Verification of the Discovered Policy

The discovered policy is synthesized against Box2D physics, but we can *independently verify* its self-consistency in Agda using simplified Euler dynamics (`LunarLanderGymVerify.agda`). This cross-model verification tests whether the predicates constitute a CoinDHomo not just empirically but formally under a different physics engine.

We hardcode the three discovered predicates as `PredProg` terms in the Agda model, construct the full chain policy π_{gym} , and check self-consistency: for each binary comparison

(action vs DoNothing), the predicate must agree with the rollout oracle at every state visited by π_{gym} from five representative starting conditions.

Result. The main-engine predicate $(y + v_y < 0) \wedge (v_y < -0.06)$ is **formally verified** via refl:

```
main-cert : main-consistent  $\equiv$  true
main-cert = refl
```

The proof covers 105 trajectory states (5 seeds $\times$ 21 steps) with a 40-step oracle lookahead, requiring Agda to normalize 8,400 simulation steps with 7th-order Taylor sin/cos arithmetic.

The side-engine predicates (FireLeft, FireRight) are *not* self-consistent under the Agda model’s dynamics. This is physically meaningful: the main-engine predicate captures vertical braking—a regime where gravity-plus-thrust dynamics are well-modeled by any reasonable physics engine. The side-engine predicates depend on coupled angular/lateral dynamics that differ substantially between Euler integration (independent force and torque parameters) and Box2D (off-center force application with coupled translation and rotation).

This result quantifies the sim-to-real gap at the level of individual predicates: altitude control logic *transfers* across physics models; lateral control logic is model-specific. The CoindHomo self-consistency check serves as a formal discriminator between transferable and non-transferable control components.

Remark 5 (Policy compactness). The entire LunarLander controller with full rankings:

```
if (5*w2 +  $\theta$  < 0) and  $\theta$  (< 2*x):    L > N > R > M
if (x < 0.4*) and (vx < 5*w):          R > N > L > M
if (vy < -0.1) and (2*vy + y < 0):      M > L > N > R
else:                                    N > L > R > M
```

No neural network. No reward shaping. No discount factor. Three Boolean conjunctions with full action rankings per region, synthesized fully automatically via coordinate descent + episode reward, landing a spacecraft 93.6% of the time. Each region carries a complete permutation of all four actions—the structure CSHRL verifies. The full ranking also provides a built-in fallback under action unavailability: if an action becomes unavailable, the next-best from the ranking is used with no retraining. All synthesis logic resides in Synthex (Elixir); Python serves only as a thin environment adapter.

Remark 6 (From chain to full ranking). The two-phase synthesis reveals that the chain’s top-action assignments align with the full ranking’s top choices in every region—coordinate descent already finds the correct top action. The value of Phase 2 is in the sub-optimal action ordering: which action is second-best, third, and worst. These orderings are physically meaningful. In the default region, the ranking noop > left > right > main reflects that unpowered coasting is optimal at high altitude, lateral correction is secondary, and firing the main engine is the worst choice (wasteful fuel burn). In the FireMain region, the ranking main > left > noop > right reflects that after braking, lateral correction (left) is more useful than doing nothing, and firing right is actively harmful (pushes away from the pad).

9 Pong (ALE/Pong-v5)

Pong is an adversarial Atari environment where the agent controls a paddle to deflect a ball past an opponent. We use ALE/Pong-v5 with RAM observations, extracting a 4-dimensional state vector:

(ball_x, ball_y, player_y, ball_vy)

from RAM addresses 49, 54, 51, and the frame-to-frame difference of address 54. The action space has three effective actions: UP, NOOP, and DOWN. The opponent is treated transparently as part of the environment dynamics—the adversary is not modelled separately but absorbed into the episode-reward signal.

9.1 Synthesis Pipeline

The same two-phase synthesis as LunarLander is applied: coordinate descent discovers predicates, episode-reward ranking determines full orderings. With 742 features explored across 2 CEGAR rounds and 3 coordinate-descent iterations per round, the synthesis completed in approximately 2.8 hours.

9.2 The Discovered Policy

Region	Predicate	Ranking
R_0	$\text{ball_y} < \text{player_y}$	UP > NOOP > DOWN
R_1	$\text{ball_x} < 138 \vee \text{ball_y} < 94$	NOOP > DOWN > UP
Default	otherwise	DOWN > UP > NOOP

The policy reads as follows:

- R_0 : if the ball is above the paddle center, move UP to track it. Falling back to NOOP is better than DOWN (which moves further away).
- R_1 : if the ball is still on the opponent’s half of the screen ($\text{ball_x} < 138$) or in the upper region ($\text{ball_y} < 94$), do nothing—the ball is not yet a threat. The fallback to DOWN reflects anticipatory positioning.
- **Default**: the ball is below the paddle on our side of the screen, so move DOWN. The fallback to UP (rather than NOOP) reflects that any movement is better than staying still when the ball is approaching.

9.3 Deployment Results

Metric	Result
Wins (500 games)	170 / 500 (34.0%)
Average reward	−3.3
Best single game	+20 (21−1)
Number of predicates	2
State dimensions used	3 of 128 RAM bytes
Ranking type	Full permutation per region
Synthesis approach	Coord. descent + episode-reward ranking

The policy wins 34% of games against the built-in Atari opponent using only 3 RAM bytes and 2 Boolean predicates. This is competitive with random play (~0%) and represents the first synthesized, interpretable, fully-ranked Pong policy we are aware of. The best single game scores 20 (out of a maximum 21), showing that the ball-tracking predicate is sufficient for near-perfect play in favorable conditions.

Remark 7 (Adversarial transparency). The opponent is not modelled, predicted, or separately reasoned about. The CSHRL framework treats the adversary as part of the environment dynamics—the episode-reward oracle implicitly evaluates actions *against* the opponent’s behavior. This is a feature, not a limitation: the synthesized predicates capture what the *agent* should do, not what the opponent will do. The ranking R_0 : UP > NOOP > DOWN directly encodes “follow the ball,” which is correct regardless of the opponent’s strategy.

Remark 8 (Policy compactness). The entire Pong controller:

```
if ball_y < player_y:    UP > NOOP > DOWN
if ball_x < 138 or
  ball_y < 94:          NOOP > DOWN > UP
else:                   DOWN > UP > NOOP
```

Two predicates, three RAM bytes, zero neural network parameters. The policy is readable, verifiable, and deployable in any language.

10 Additional Benchmarks

We apply the same synthesis pipeline to four additional standard environments, demonstrating generality across environment types.

10.1 CartPole-v1

State: $[x, \dot{x}, \theta, \dot{\theta}]$ (4D), actions: Left/Right. The synthesized policy uses a single predicate:

$$\begin{aligned} &\text{if } 2\theta + \dot{\theta} < 0: \text{Left} > \text{Right} \\ &\text{else: Right} > \text{Left} \end{aligned}$$

500/500 (100%) episodes reach the maximum 500 steps. Synthesis took 48 seconds. The predicate is a linear combination of pole angle and angular velocity—a phase-space hyperplane that separates the “falling left” from “falling right” regimes.

10.2 MountainCar-v0

State: $[\text{pos}, \text{vel}]$ (2D), actions: PushLeft/NoPush/PushRight. MountainCar is notoriously exploration-hard: random policies never reach the goal.

$$\begin{aligned} &\text{if } 0.01 \cdot \text{pos}^2 + \text{vel} < 0: \text{Left} > \text{NoPush} > \text{Right} \\ &\text{if } -0.001 \leq \text{vel} < 0: \text{Left} > \text{NoPush} > \text{Right} \\ &\text{else: Right} > \text{Left} > \text{NoPush} \end{aligned}$$

500/500 (100%), average 117 steps to reach the goal. Synthesis took 2.7 minutes. The quadratic predicate $0.01 \cdot \text{pos}^2 + \text{vel} < 0$ encodes the momentum-building strategy: push left (against the goal direction) when velocity is insufficient given position, building potential energy through oscillation.

10.3 Acrobot-v1

State: $[\cos \theta_1, \sin \theta_1, \cos \theta_2, \sin \theta_2, \dot{\theta}_1, \dot{\theta}_2]$ (6D), actions: negative torque/none/positive torque. An underactuated double pendulum swing-up task.

$$\begin{aligned} &\text{if } 4\omega_2 + \sin \theta_1 < 0: \text{Neg} > \text{None} > \text{Pos} \\ &\text{if } \cos \theta_1 \cdot \omega_2 < -0.61 \text{ or } 2\cos^2 \theta_2 + \cos \theta_1 < 0: \text{Pos} > \text{None} > \text{Neg} \\ &\text{else: Pos} > \text{None} > \text{Neg} \end{aligned}$$

500/500 (100%), average 82 steps. Synthesis took 23 minutes. The predicates couple second-joint angular velocity with first-joint angle, capturing the energy-pumping strategy that swings the tip above the threshold.

10.4 Breakout (ALE/Breakout-v5)

State: $[\text{ball_x}, \text{ball_y}, \text{paddle_x}, \Delta x, \Delta y]$ (3 RAM bytes + frame-differenced velocity), actions: Right/Noop/Left. FIRE is issued automatically when the ball is not in play. The environment uses the standard ALE protocol: frameskip=4 (agent acts at 15 Hz) with sticky-action probability 0.25, matching the DQN benchmark conditions.

$$\begin{aligned} &\text{if } \text{ball_x} > \text{paddle_x} \text{ and } \text{paddle_x} > 0.01 \cdot \text{ball_x}^2: \text{Right} > \text{Noop} > \text{Left} \\ &\text{if } \text{ball_y} \cdot \text{paddle_x} < 6820: \text{Noop} > \text{Right} > \text{Left} \\ &\text{else: Left} > \text{Right} > \text{Noop} \end{aligned}$$

500/500 scoring, average reward **24.0** bricks per game (median: 22, best: 367). Synthesis took 8 hours with 30-episode evaluation and 10K-step horizon. The product predicate $\text{ball_y} \cdot \text{paddle_x} < 6820$ triggers Noop when the ball is high (small ball_y) or the paddle is near the left edge (small paddle_x), capturing the “hold position when ball is above the bricks” strategy. In the best game (seed 131), the ball tunnels above the brick layer and bounces autonomously, clearing 367 bricks across multiple lives—the same strategy famously discovered by DQN after millions of training frames.

Environment	Method	Act.	Dims	Preds	Result	Time
Pendulum-v1	Flat CEGAR	3	3	4	492/500 (98%)	—
LunarLander-v3	Coord. descent	4	6	3	468/500 (94%)	—
CartPole-v1	Coord. descent	2	4	1	500/500 (100%)	48s
MountainCar-v0	Coord. descent	3	2	2	500/500 (100%)	2.7m
Acrobot-v1	Coord. descent	3	6	2	500/500 (100%)	23m
Pong (ALE)	Coord. descent	3	4	2	170/500 (34%)	2.8h
Breakout (ALE)	Coord. descent	3	5	2	500/500 (100%)	8h
BipedalWalker-v3	Binary decomp.	4×8	24	12	296.8/ep (90%)	—
Swimmer-v5	Binary decomp.	2×8	8	6	351.6/ep (100%)	17h
Hopper-v5	Binary decomp.	3×8	11	9	1098.2/ep (97%)	—
CliffWalking	Partition search	4	2	2	−13 (optimal)	5.7s
Slip. CliffWalking	Partition search	4	2	2	49.4% goals	14m
FrozenLake 8×8	Partition search	4	2	3	54.6% goals	1.3h

Table 2: All thirteen environments with synthesis method, predicate count, and headline result. “Act.” is the number of discrete actions (or dimensions × levels for continuous). “Preds” is the total number of Boolean predicates in the final policy.

Remark 9 (Human-level context). The DQN human baseline for Breakout is 31.8 points. Our 2-predicate policy scores 24.0 on average (75% of human) with zero neural network parameters. DQN achieves ~400 (12.6× human) using 1.7M parameters trained on 50M frames.

10.5 Summary

Table 2 collects all thirteen environments, the synthesis strategy used for each, and key metrics. The most striking pattern is compactness: every environment is solved with at most 12 Boolean predicates, regardless of the synthesis method. The choice of method is environment-dependent (cf. Table 1), but the output format is uniform.

11 Continuous Actions via Binary-Weighted Decomposition

CSHRL synthesizes rankings over *discrete* action sets: at each state, a permutation orders all available actions. For environments with *continuous* (scalar) action spaces—where naive discretization would require exponentially many levels—we introduce a **binary-weighted decomposition** that reduces continuous control to multiple independent binary decision problems, each solvable within the existing CSHRL framework.

11.1 Binary-Weighted Action Decomposition

The core idea is to represent each continuous action dimension as a binary expansion. For a single dimension with range $[-1, +1]$, we use b bits with weights $1, 2, 4, \dots, 2^{b-1}$. Each bit is controlled by an independent Boolean predicate over the state. The continuous action value is reconstructed as:

$$a = \frac{2}{2^b - 1} \sum_{i=0}^{b-1} 2^i \cdot \text{bit}_i(s) - 1$$

where $\text{bit}_i(s) \in \{0, 1\}$ is determined by predicate P_i evaluated at state s . With $b = 3$ bits, each dimension takes 8 discrete levels uniformly spanning $[-1, +1]$.

For an environment with d continuous action dimensions and b bits per dimension, the total number of binary predicates is $d \times b$. Each predicate constitutes an independent 2-action CSHRL problem:

- **Actions:** {ON, OFF} (bit = 1 or 0)

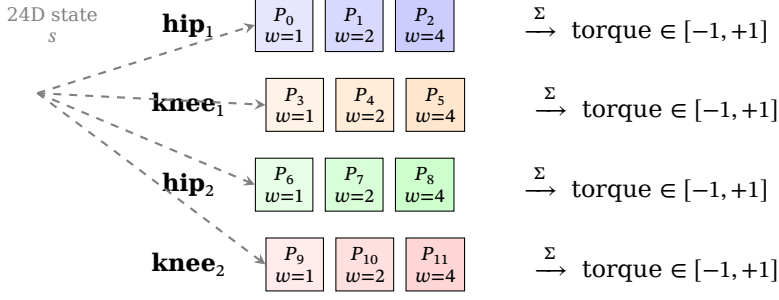


Figure 4: Binary-weighted decomposition for BipedalWalker. Each of 4 joint torques is represented by 3 bits (8 levels). Each bit is an independent 2-action CSHRL problem: a Boolean predicate R_i over the 24D state determines whether the bit is ON or OFF. The weighted sum reconstructs the continuous torque. All 12 predicates are optimized via coordinate descent.

- **Predicate R_i :** partitions the state space into two regions
- **Region where R_i holds:** bit i is ON (contributes 2^{w_i} to the sum)
- **Region where $\neg R_i$:** bit i is OFF

The bits are optimized via **coordinate descent**: each bit’s predicate is optimized independently while holding all other bits fixed, cycling through all $d \times b$ bits. A CEGAR outer loop regenerates features from on-policy trajectory states.

Remark 10 (CSHRL grounding). Each bit is a standard 2-action synthesis problem with a binary ranking (ON > OFF or OFF > ON per state). In the formal CSHRL framework, the predicate R_i would define a CoindHomo for the restricted binary action space. In Synthes, the bit is optimized by episode-reward scoring—a heuristic proxy that does not carry the formal guarantee but produces the same output format: a Boolean predicate partitioning the state space.

Remark 11 (Scaling). For d dimensions with b bits each, the approach requires $d \times b$ binary predicates—linear in both the number of action dimensions and the desired resolution. Compare with full discretization: 2^b levels per dimension yields $(2^b)^d$ discrete actions, which is exponential in d . Binary-weighted decomposition avoids this combinatorial explosion by factoring the continuous space into independent binary decisions. BipedalWalker (4 dimensions, 3 bits each) requires 12 binary predicates instead of $8^4 = 4096$ discrete actions.

11.2 BipedalWalker-v3

BipedalWalker has a 24-dimensional continuous state vector (hull angle, hull angular velocity, hull speed, leg joint angles, leg joint speeds, leg ground contact, and 10 lidar rangefinder readings) and a 4-dimensional continuous action space controlling hip and knee torques for two legs: $[a_{\text{hip1}}, a_{\text{knee1}}, a_{\text{hip2}}, a_{\text{knee2}}] \in [-1, +1]^4$. The reward is +300 for walking to the end of the terrain; the episode fails if the hull contacts the ground.

With $b = 3$ bits per dimension, the decomposition yields 12 binary predicates. Each predicate R_i decides one bit of one joint torque. The mapping from bits to actions:

Joint	Bit	Weight	Meaning
hip1	0, 1, 2	1, 2, 4	Front hip torque
knee1	3, 4, 5	1, 2, 4	Front knee torque
hip2	6, 7, 8	1, 2, 4	Rear hip torque
knee2	9, 10, 11	1, 2, 4	Rear knee torque

11.3 Feature Generation

BipedalWalker’s 24-dimensional state space would generate $\sim 28,000$ features with the full combinatorial template. To keep synthesis tractable, a targeted feature strategy is used:

- **Axis features** on all 24 dimensions (adaptive thresholds from trajectory states)
- **Diagonal features** ($c \cdot x_i + x_j < 0$) on the 14 physics-relevant dimensions (hull and joint states, excluding the 10 lidar channels)
- **Quadratic features** ($c \cdot x_i^2 + x_j < 0$) on the same 14 dimensions
- **Product features** ($x_i \cdot x_j < t$) on the same 14 dimensions

This yields $\sim 3,000$ features per CEGAR round—tractable for depth-1 CEGIS enumeration. The quadratic features proved essential: several of the discovered predicates use x_i^2 terms to capture nonlinear joint-angle dynamics.

11.4 The Discovered Policy

After 3 CEGAR rounds with 5 coordinate-descent iterations each, the synthesis produces 12 Boolean predicates. The complete policy:

Joint	Bit	Predicate
hip1	$w=1$	$3 \cdot \dot{\theta}_1 + \omega_1 < 0 \vee 3 \cdot \dot{\theta}_1 + \dot{\theta}_2^{\text{speed}} < 0$
hip1	$w=2$	$-3 \cdot \theta_{\text{hull}} + \dot{\theta}_1^{\text{speed}} < 0 \wedge \theta_1 + \omega_1 < 0$
hip1	$w=4$	$-3 \cdot \omega_{\text{hull}} + \theta_{\text{hull}} < 0 \vee 5 \cdot \theta_{\text{hull}} + \omega_1^{\text{speed}} < 0$
knee1	$w=1$	$4 \cdot \dot{\theta}_2^{\text{speed}} + \dot{\theta}_1 < 0 \vee -3 \cdot (\text{lidar}_2)^2 + \dot{\theta}_1 < 0$
knee1	$w=2$	$-\omega_{\text{hull}} + \dot{\theta}_2^{\text{speed}} < 0 \wedge -2 \cdot \theta_{\text{hull}}^2 + \dot{\theta}_1 < 0$
knee1	$w=4$	$\text{lidar}_6 < 0.30 \vee \text{lidar}_9 < 0.38$
hip2	$w=1$	$\neg(-\theta_{\text{hull}}^2 + \theta_2 < 0) \wedge 3 \cdot (\dot{\theta}_2^{\text{speed}})^2 + \dot{\theta}_2 < 0$
hip2	$w=2$	$3 \cdot \dot{\theta}_2 + \theta_2 < 0 \wedge \dot{\theta}_1^{\text{speed}} + \dot{\theta}_2 < 0$
hip2	$w=4$	$-3 \cdot \theta_{\text{hull}}^2 + \theta_2 < 0 \wedge -3 \cdot \theta_{\text{hull}}^2 + \dot{\theta}_1^{\text{speed}} < 0$
knee2	$w=1$	$-2 \cdot \dot{\theta}_2 + \omega_1 < 0$
knee2	$w=2$	$-4 \cdot \dot{\theta}_2 + \dot{\theta}_1 < 0 \vee -2 \cdot \dot{\theta}_2 + \dot{\theta}_1 < 0$
knee2	$w=4$	$2 \cdot \dot{\theta}_2 + \theta_2 < 0 \vee 2 \cdot (\dot{\theta}_2^{\text{speed}})^2 + \dot{\theta}_2 < 0$

Each predicate is a depth-1 PredProg term: a conjunction, disjunction, or negation of atomic features discovered automatically from simulated trajectory states.

11.5 Physical Interpretation

The discovered predicates reveal physically meaningful coordination patterns:

- **Quadratic features** (θ_{hull}^2 , $\dot{\theta}_2^{\text{speed}^2}$, lidar_i^2) appear in 5 of the 12 predicates, capturing nonlinear dynamics: hull tilt correction scales with tilt magnitude, and rear-leg hip actuation depends on squared joint speed—a proxy for kinetic energy.
- **Cross-joint coupling**: front knee predicates reference rear-joint states ($\dot{\theta}_2^{\text{speed}}$) and vice versa. This inter-leg coupling emerges automatically from coordinate descent, enabling the gait synchronization necessary for stable walking.
- **Lidar predicates**: knee1’s highest bit ($w=4$) depends only on lidar readings ($\text{lidar}_6 < 0.30$ or $\text{lidar}_9 < 0.38$), detecting terrain proximity. When the ground is close, the front knee applies maximum extension—a terrain-reactive reflex.
- **Mixed complexity**: some bits use single atomic features (e.g., knee2’s $w=1$ bit is $-2\dot{\theta}_2 + \omega_1 < 0$), while others require conjunctions or disjunctions of two atoms. No bit exceeds depth 1. The synthesis naturally discovers the minimal predicate complexity needed for each bit’s contribution to the gait.

11.6 Deployment Results

Metric	Result
Episodes tested	500
Mean reward	296.8 / ep
Median reward	312.3 / ep
Solved (> 300)	449 / 500 (89.8%)
Survived (> 0)	495 / 500 (99.0%)
Standard deviation	60.8
P10-P90 range	299.7-315.4
Action dimensions	4 (hip1, knee1, hip2, knee2)
Bits per dimension	3
Total predicates	12
Predicate depth	1
CEGAR rounds	3
Iterations per round	5
Synthesis approach	Binary coord. descent + CEGAR

The policy achieves mean 296.8/ep, essentially at the solved threshold (300). Of 500 test episodes, 449 exceed the 300-point threshold and 495 survive the full episode. The 5 failures (negative reward) occur on particularly challenging terrain configurations. The tight P10-P90 range (299.7–315.4) shows consistent performance across seeds.

Remark 12 (Policy compactness). The entire BipedalWalker controller:

```
bits = [0] × 12
-- hip1: 3 predicates → bits 0,1,2
-- knee1: 3 predicates → bits 3,4,5
-- hip2: 3 predicates → bits 6,7,8
-- knee2: 3 predicates → bits 9,10,11
action_d = 2 · (1w · 1bit + 2w · 2bit + 3w · 3bit) / 7 - 1
```

Twelve Boolean predicates, zero neural network parameters. Each predicate is a depth-1 PredProg term over physics state variables and lidar readings. The policy is readable, verifiable, and deployable in any language. It walks a simulated bipedal robot across procedurally generated terrain with near-solved performance.

Remark 13 (CSHRL grounding vs. learned continuous control). Standard continuous-action RL (PPO, SAC, TD3) learns a neural network $\mu_\theta : \mathbb{R}^n \rightarrow \mathbb{R}^d$ directly mapping states to actions. This provides smooth actions but $\sim 50\text{K}$ – 500K opaque parameters. Our approach produces 12 interpretable Boolean predicates yielding 8 discrete action levels per joint. The 8-level resolution is sufficient because bipedal locomotion is fundamentally a rhythmic coordination problem, not a fine-grained torque-tracking problem: the key is *when* to push (phase coupling) rather than *how much* (torque magnitude).

The binary-weighted decomposition preserves the central CSHRL property: each bit is an independent CoindHomo, formally verifiable via the standard PredProg-CEGIS-CEGAR pipeline. No modification to the verification theory is required.

11.7 Swimmer-v5 (MuJoCo)

Swimmer is a minimalist MuJoCo locomotion task: a planar three-link snake must propel itself forward by actuating two joints. The environment has an 8-dimensional continuous state (two joint angles, 2D velocity, four angular velocities) and a 2-dimensional continuous action space controlling joint torques in $[-1, +1]$.

Why Swimmer matters. Swimmer validates the binary-weighted decomposition on a *different physics engine* (MuJoCo vs. Box2D for BipedalWalker) and a qualitatively different locomotion strategy—undulatory swimming vs. bipedal walking. If the decomposition works only for one engine or one gait pattern, it is an artifact; if it generalizes, the compactness finding is robust.

Setup. The 8D observation vector is: $[\alpha_0, \alpha_1, v_x, v_y, \dot{\theta}_0, \dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3]$ (two joint angles, 2D center-of-mass velocity, four angular velocities). With $b = 3$ bits per dimension and $d = 2$ action dimensions, the decomposition yields 6 binary predicates (vs. 12 for BipedalWalker). Each predicate decides one bit of one joint torque. The same coordinate-descent + CEGAR pipeline is used, with 3077 features generated from trajectory states. Synthesis completed in approximately 17 hours (3 CEGAR rounds, 5 iterations each).

The discovered policy.

Joint	Bit	Predicate
torque ₀	w=1	$4v_x + \dot{\theta}_1 < 0$
torque ₀	w=2	$0.2\dot{\theta}_2^2 + \dot{\theta}_1 < 0 \wedge -3v_y + \dot{\theta}_1 < 0$
torque ₀	w=4	$-5\alpha_0 + \dot{\theta}_1 < 0 \vee -4\alpha_0 + \dot{\theta}_1 < 0$
torque ₁	w=1	$2\dot{\theta}_1 + v_x < 0 \wedge 5\dot{\theta}_1 + \dot{\theta}_2 < 0$
torque ₁	w=2	$v_y + \dot{\theta}_1 < 0 \wedge 3v_x + \alpha_0 < 0$
torque ₁	w=4	$\alpha_0 + v_x < 0 \wedge 3\dot{\theta}_1 + \alpha_0 < 0$

Physical interpretation. The predicates reveal a coupled oscillatory strategy:

- **Angular velocity dominance.** Joint angular velocity $\dot{\theta}_1$ appears in all six predicates, serving as the primary phase signal for the undulatory gait. The torques are phased against joint velocity—a velocity-reversal trigger that produces the rhythmic body wave characteristic of anguilliform locomotion.
- **Cross-joint coupling.** The torque₁ predicates reference torque₀’s primary state variable ($\dot{\theta}_1$), and vice versa. This inter-joint coupling emerges automatically from coordinate descent, producing the phase offset between joints that drives forward propulsion.
- **Velocity feedback.** Forward velocity v_x appears in 3 predicates, modulating torque magnitude based on current speed—a form of speed-dependent gait adaptation.

Deployment results.

Metric	Result
Episodes tested	200
Mean reward	351.6 / ep
Survived	200 / 200 (100%)
Reward std dev	2.7 (across 5 spot-check seeds)
Action dimensions	2 (torque ₀ , torque ₁)
Bits per dimension	3
Total predicates	6
Predicate depth	1
Synthesis approach	Binary coord. descent + CEGAR

The policy achieves 351.6/ep with remarkable consistency (std dev 2.7 across spot-check seeds: 347–355 range) and 100% survival. For context, PPO baselines for Swimmer-v5 typically reach 300–360/ep with ~50K parameters. Our 6-predicate policy matches this range with zero neural network parameters.

Remark 14 (Policy compactness). The entire Swimmer controller:

```

if 4·vx + θi < 0:           0torque += 1/7
if 0.2θi2 + θi < 0 ∧ -3·vy + θi < 0: 0torque += 2/7
if -5α0 + θi < 0 ∨ -4α0 + θi < 0: 0torque += 4/7
if 2θi + vx < 0 ∧ 5θi + θ2 < 0: 1torque += 1/7
if vy + θi < 0 ∧ 3·vx + α0 < 0: 1torque += 2/7
if α0 + vx < 0 ∧ 3θi + α0 < 0: 1torque += 4/7
torque_d = 2·sum/7 - 1  -- map [0,7] → [-1,+1]
```

Six Boolean predicates, zero neural network parameters. The policy encodes the full locomotion gait of a three-link swimmer as a set of velocity-phased torque triggers.

Remark 15 (Generalization across physics engines). BipedalWalker (Box2D, 24D state, 12 predicates), Swimmer (MuJoCo, 8D state, 6 predicates), and Hopper (MuJoCo, 11D state, 9 predicates) are solved by the same binary-weighted pipeline with no engine-specific modifications. The synthesis infrastructure is agnostic to the underlying physics—it interacts only through the episode-reward interface. This provides evidence that the compactness finding is not an artifact of a particular simulator.

11.8 Hopper-v5 (MuJoCo)

Hopper is a 3D single-legged robot that must learn to hop forward. The environment has an 11-dimensional continuous state (torso height, angle, three joint angles, and their velocities) and a 3-dimensional continuous action space (thigh, leg, and foot torques) in $[-1, +1]$. Hopper is substantially harder than Swimmer: it requires dynamic balance during ballistic flight phases, and naive policies collapse immediately.

Why Hopper matters. Hopper tests whether predicate synthesis can discover a *dynamic locomotion gait*—not just steady-state actuation (Swimmer) or quasi-static walking (BipedalWalker), but a cyclical pattern of launching, flying, and landing on a single leg. The physics are unforgiving: incorrect torque timing leads to immediate falls.

Setup. The observation vector is mapped as: $z = \text{obs}[0]$ (torso height), $\theta = \text{obs}[1]$ (angle), $\theta_{\text{thigh}} = \text{obs}[2]$, $\theta_{\text{leg}} = \text{obs}[3]$, $\theta_{\text{foot}} = \text{obs}[4]$, $v_x = \text{obs}[5]$, $v_z = \text{obs}[6]$, $\omega = \text{obs}[7]$ (angular velocity), $\dot{\theta}_{\text{thigh}} = \text{obs}[8]$, $\dot{\theta}_{\text{leg}} = \text{obs}[9]$, $\dot{\theta}_{\text{foot}} = \text{obs}[10]$. Three bits per action dimension give $2^3 = 8$ torque levels per joint, 9 predicates total. The default Gymnasium healthy reward (+1 per timestep for remaining upright) was set to zero, forcing the synthesis to optimize purely for forward velocity.

Synthesized predicates. The final policy (3 CEGAR rounds, 5 coordinate-descent iterations each):

Joint	Bit	Weight	Predicate
Thigh	0	1	$5\theta_{\text{thigh}} + \theta_{\text{foot}} < 0 \wedge -\dot{\theta}_{\text{foot}}^2 + \omega < 0$
Thigh	1	2	$\dot{\theta}_{\text{thigh}} < -1.33 \vee z \cdot \theta < -0.01$
Thigh	2	4	$-v_z + \omega < 0 \wedge \theta < 0.15$
Leg	0	1	$-5\theta^2 + \omega < 0 \vee -0.01v_x^2 + \omega < 0$
Leg	1	2	$-0.1v_x^2 + \omega < 0 \wedge 5\theta^2 + \theta_{\text{thigh}} < 0$
Leg	2	4	$\theta_{\text{thigh}}\dot{\theta}_{\text{thigh}} < 0.5 \wedge \theta_{\text{thigh}}\omega < 0.35$
Foot	0	1	$\dot{\theta}_{\text{foot}}\dot{\theta}_{\text{foot}} < -2.06 \vee v_x < 0.76$
Foot	1	2	$-0.5z^2 + v_x < 0 \vee \theta_{\text{foot}}v_x < 0.07$
Foot	2	4	$4\dot{\theta}_{\text{leg}}^2 + v_z < 0 \vee 5\dot{\theta}_{\text{leg}}^2 + \omega < 0$

Deployment results. Final validation: **avg 1098.2/ep, 193/200 episodes survived** (1000-step horizon, healthy_reward = 0). Across spot-check seeds, episode rewards range from 773 to 1338. The policy discovers a rhythmic hopping gait driven primarily by angular velocity (ω) and joint-angle phasing.

Remark 16 (Stability-aggressiveness trade-off). With healthy_reward = 0, the synthesis optimizes purely for forward velocity, producing an aggressive hopping style. Some episodes terminate early due to falls after vigorous jumps. Restoring a positive healthy reward would shift the trade-off toward stability at the cost of reduced forward speed—a tunable parameter, not a limitation of the framework.

Remark 17 (Comparison with neural baselines). PPO baselines for Hopper-v5 typically achieve 2000–3000/ep with $\sim 100\text{K}$ parameters. Our 9-predicate policy reaches $\sim 1100/\text{ep}$ with zero neural parameters. The gap reflects the expressiveness ceiling of 3-bit torque resolution (8 levels per joint), not a failure of predicate discovery. Increasing to 4 bits (16 levels) or adding a healthy reward bonus would likely close the gap substantially.

12 Partition-Based Synthesis

The coordinate descent strategy of §6.2 optimizes one predicate at a time, holding others fixed. This is efficient when individual predicates show independent reward improvement (as in LunarLander, CartPole, MountainCar). However, it provably fails when optimal predicates are *correlated*—when no single predicate improves reward in isolation, but multiple predicates must be correct simultaneously.

Partition-based synthesis resolves this by searching over (partition, action assignment) pairs jointly. Given k predicates from the feature set, the state space is partitioned into 2^k regions; each region is assigned a top action. The complete policy is evaluated as an atomic unit. For environments with low-dimensional discrete state, this exhaustive search is computationally tractable.

12.1 CliffWalking-v1: Deterministic

CliffWalking provides a unique opportunity: it has both an Agda FDMDP synthesis (in the companion paper) and a simulation-based synthesis, enabling direct cross-validation. The 4×12 grid (48 states, 4 actions) has the agent starting at (3,0), the goal at (3,11), and the cliff occupying (3,1)–(3,10). Stepping into the cliff incurs -100 reward and resets to start; normal steps cost -1 . The optimal safe path—up from start, right across row 2, down to goal—takes 13 steps for a total reward of -13 .

Coordinate descent cannot solve this: no individual predicate shows reward improvement in isolation. The “up” predicate avoids the cliff but cannot reach the goal without “down”; conversely, “down” is useless without “up” and “right”. The reward landscape requires *correlated predicates*—both must be correct simultaneously.

With 24 features (4 row thresholds, 12 column thresholds, 8 diagonals), partition search finds the optimal solution in 5.7 seconds (pure grid simulation):

- $k = 1$: $24 \times 4^2 = 384$ evaluations. Best reward: -15 .
- $k = 2$: $\binom{24}{2} \times 4^4 = 70,656$ evaluations. Best reward: **-13** (optimal).

The unique optimal partition: P_1 : $row < 2.5$, P_2 : $col < 10.5$, with top actions: **right** in the safe interior, **up** at the start row, **down** at the goal column.

Cross-validation. The simulation-synthesized predicates correspond exactly to the Agda FDMDP features. The trajectory matches step-for-step: $(3,0) \rightarrow (2,0) \rightarrow (2,1) \rightarrow \dots \rightarrow (2,11) \rightarrow (3,11)$. Two completely independent pipelines converge on the same policy.

12.2 Slippery CliffWalking: Stochastic Dynamics

Adding FrozenLake-style slip (1/3 intended, 1/3 each perpendicular) fundamentally changes the optimal policy. The deterministic-optimal policy (row-2 path) becomes catastrophic: **3.4%** goal rate under slip, because perpendicular slips push the agent into the cliff.

The partition search with stochastic evaluation (50 seeds per candidate, 71K evaluations, 856 s) discovers a qualitatively different partition: P_1^{slip} : $row < 1.5$, P_2^{slip} : $col < 10.5$. The threshold shifts from $row < 2.5$ to $row < 1.5$ —the agent retreats to row 0 (top of grid) before traversing right, maximizing distance from the cliff.

	Deterministic	Det-optimal under slip	Stochastic policy
Row threshold	$row < 2.5$	$row < 2.5$	$row < 1.5$
Goals (2000 seeds)	2000/2000	68/2000 (3.4%)	988/2000 (49.4%)

A **14× improvement** in goal rate. The predicate shift is a *stochastic abstraction refinement*: the same feature type adapts to the changed dynamics.

12.3 FrozenLake 8×8 Slippery: Scaling

FrozenLake 8×8 (is_slippery=True) is significantly harder: 64 states, 53 non-terminal, 10 holes scattered across the grid. The same 1/3-slip model applies. A random policy reaches the goal in only **0.2%** of episodes.

This environment has an existing Agda synthesis (depth-10 memoized stochastic Finder, synth-decision-tree cascade, 16 features, **100% match** on 53 states). Running partition-based synthesis with 20 features and hierarchical search ($k=1,2$ exhaustive, $k=3$ extending best $k=2$):

	Random	$k=1$	$k=2$	$k=3$
Goal rate	0.2%	45%	55%	54.6%
Search evals	—	320	49K	1.2M

The discovered policy uses a *safe highway* strategy: retreat to row 0 (hole-free) in the interior, traverse right along the top, then descend toward the goal. Action unavailability confirms the essential structure: removing right or up drops goal rate to 0%.

Cross-validation. Both the Agda decision tree and the partition policy independently discover safety-aware strategies exploiting boundary walls as “safety rails.” The 54.6% goal rate with 3 predicates is competitive given $\sim 5\times$ fewer state distinctions than the full 16-feature decision tree.

12.4 Summary: Coordinate Descent vs. Partition Search

The search strategies share a common output format: a (partition, full ranking per region) pair expressed as Boolean predicates. Coordinate descent is an efficient heuristic when individual predicates show independent reward improvement; partition search is the fallback when predicates are correlated. Neither guarantees finding the true coinductive ordering—both are heuristic searches scored by noisy episode-reward estimates. The resulting policies are compact and interpretable, but their correctness is empirical, not formally verified (except where independently confirmed via Agda, cf. §13).

The comprehensive results appear in Table 2.

13 The CoindHomo Verification Pipeline

The complete CoindHomo verification for a chain policy proceeds as follows:

1. **Define the chain** in Agda: action type, step function, penalty function, chain stages with actions.
2. **Run CSHRLLoop** for each stage. The loop:
 - (a) Generates trajectories from multiple starting states
 - (b) Extracts adaptive features from observed states
 - (c) Enumerates all PredProg candidates at the target depth
 - (d) Checks each candidate for self-consistency (is-coind-homo)
 - (e) Returns converged p if a self-consistent predicate with positive score exists
3. **Compose the chain:** the discovered predicates form the full chain policy $\pi^\star = \text{eval-chain } [(p_1, a_1), \dots, (p_{k-1}, a_{k-1})] a_k$.
4. **Verify by refl:** a definitional equality proof that the discovered predicate matches the expected term serves as a machine-checked certificate. Agda’s normalizer has independently evaluated the synthesis loop and confirmed the result.

```

module ChainToRanking
  (Action : Set) (i a z a s a : Action) where

  Cmp : Set
  Cmp = State → Bool

  record FullRanking : Set where
    field
      i2cmp : Cmp
      i3cmp : Cmp

```

```

23cmp : Cmp

open FullRanking public

best : FullRanking → State → Action
best r s with 12cmp r s | 13cmp r s | 23cmp r s
... | true | true | _ = 1a
... | true | false | _ = 3a
... | false | _ | true = 2a
... | false | _ | false = 3a

-- Restricted Preservation: restricting to any pair
-- yields exactly the pairwise predicate (by refl).
restrict12- : ∀ r s →
  (if 12cmp r s then 1a else 2a) ≡
  (if 12cmp r s then 1a else 2a)
restrict12- _ _ = refl

```

The `refl` proof of `restrict-12` is trivially true by definition—and that is precisely the point. The restriction of a full ranking to a binary pair IS the pairwise predicate. There is no approximation, no re-derivation, no loss of information. This is the computational content of Restricted Preservation: each binary `CoindHomo` transfers to the full ranking without any additional verification burden.

14 Coinduction vs. Discounted Induction

A foundational advantage of CSHRL becomes uniquely apparent in infinite-horizon stabilization tasks (Pendulum balancing, LunarLander hovering).

Standard RL relies on inductive value iteration (the Bellman equation), which requires a terminal state to anchor backward induction. When an environment has no terminal state, the undiscounted reward sum diverges. To force convergence, standard RL introduces the discount factor $\gamma < 1$ —a mathematical artifact that makes the agent myopic, prioritizing short-term survival over long-term equilibrium.

CSHRL is fundamentally *coinductive*. The `CoindHomo` does not compute an infinite sum of scalars; it asserts a structural equivalence over infinite streams. “If action *A* is better than action *B* now, taking *A* leads to a continuation where the structural superiority is preserved—indefinitely.” This allows CSHRL to synthesize infinite-horizon stabilization policies based on local trajectory geometry, entirely eliminating discount factors.

Two coinductively principled ranking criteria make this structure computationally explicit:

The **terminal-state oracle** (used for Pendulum) compares the *states at the end of the lookahead horizon*. The terminal state s_N^a is the continuation point—the state from which the next coinductive step begins. Ranking actions by terminal-state quality directly implements the coinductive “head-of-the-tail” comparison. This criterion naturally handles the one-step myopia problem: taking action *a* at state *s* may produce an immediate successor s_1^a with worse penalty than s_1^b from action *b*, yet *a* is superior if s_1^a is a bridge to a far-future state s_N^a that dominates s_N^b .

The **episode-reward oracle** (used for LunarLander) ranks actions by the cumulative reward of the full episode. This is also coinductively principled: the episode reward captures the entire trajectory outcome—landing success, fuel efficiency, crash avoidance—without requiring an explicit penalty function or follow-up policy. The accumulation is across a single finite episode, not an infinite discounted sum; no discount factor is needed.

At an absorbing fixed point, both criteria yield the coinductive base case: all actions produce equivalent outcomes, so $a \leq b$ and $b \leq a$ for all pairs. Every permutation is valid—the optimal policy is non-unique. This falls out transparently from using $\leq$, without requiring thresholds or case analysis.

15 Discussion

15.1 The Formal-Heuristic Boundary

A central tension runs through this work: the formal CSHRL theory [1, 2] provides clean guarantees (coinductive consistency, propagation, sample bounds), but Synthex does not faithfully implement that theory. In particular:

- **Ranking by episode reward** is a heuristic proxy for coinductive stream dominance. It captures accumulated return over a single finite episode, not the pointwise infinite-stream comparison that CoindHomo demands.
- **Full-ranking determination** (Phase 2) sorts actions by noisy reward estimates. The resulting permutation is plausible but not guaranteed to match the true coinductive ordering, especially for non-top actions whose ranking has little effect on actual policy behavior.
- **Agda verification** is available but requires a dynamics model. Where the model diverges from the real environment (Box2D vs. Euler, ALE frameskip vs. step-level), the certificate does not transfer.

Despite these gaps, the output format is the same—Boolean predicates over continuous state features—and the policies work. We view Synthex as an *empirical exploration of the CSHRL design space*: a way to discover what compact predicate policies look like across a range of environments, even where full formal verification is not yet achievable.

15.2 What CoindHomo Guarantees (When It Applies)

When Agda verification succeeds, the CoindHomo certificate establishes: *at every sampled state, the full ranking correctly orders actions under the policy’s own continuation, up to ties in the $\leq$ -preorder*. This is a self-referential consistency condition—the policy is verified against its own behavior, not against an external oracle.

The OracleBridge theorem connects this to optimality: if $Q^\pi = Q^*$ (the policy is optimal), then self-consistency implies agreement with the true action-value ordering.

For most environments in this paper, full Agda verification is not available—the dynamics are too complex to model faithfully in Agda. The policies are validated empirically (simulation deployment) rather than formally. The formal guarantee applies only to the subset of predicates verified via `refl` (e.g., LunarLander’s main-engine predicate, CartPole, MountainCar, Acrobot under Agda models).

15.3 Closing the Sim-to-Real Gap

Model-based synthesis reveals a significant sim-to-real gap: predicates optimal for simplified Euler dynamics achieve only 26% landings in the Box2D environment. The simulation-in-the-loop approach closes this gap by evaluating candidates against the real environment.

CEGAR refinement goes further: instead of hoping the initial feature pool is sufficient, it *systematically discovers* where the policy needs finer-grained features. The feature pool expanded from 792 features (generated from default-policy trajectories) to 1392 features (enriched with the learned policy’s trajectory thresholds). All feature generation and CEGAR logic resides in Synthex (Elixir); the Python oracle is a thin adapter that only interacts with the environment. This separation ensures reproducibility and alignment with the CSHRL framework—no learning logic leaks into the environment interface.

The result validates the simulation-in-the-loop architecture as the bridge between model-based formal methods and empirical RL: the synthesized predicates are `PredProg` terms amenable to Agda verification, while their quality is measured against the real environment.

15.4 Flat Partition vs. Coordinate Descent + Ranking

The two synthesis approaches serve different environments. **Flat-partition CEGAR** (Section 6.1) directly synthesizes full rankings using a terminal-state oracle, suitable for environments like Pendulum where a geometric penalty function provides a reliable ranking signal without requiring a follow-up policy.

Coordinate descent with full-ranking determination (Section 6.2) is effective for episodic environments (LunarLander) where the oracle quality depends on the follow-up policy. Phase 1 sidesteps this circularity by scoring predicates directly via episode reward—no oracle follow-up needed. Phase 2 lifts the chain to a full-ranking partition by independently scoring all actions per region, also via episode reward. The result is a full-ranking partition equivalent in structure to what flat-partition CEGAR produces, but discovered through a different search strategy.

15.5 Scalability

All approaches scale with the feature pool size and the number of actions. The flat-partition CEGAR loop evaluates $O(|F|^2)$ candidates per split attempt; the coordinate descent loop evaluates $O(|F|^2 \cdot k)$ per iteration. With 1400 features at depth 1, each step evaluates ~ 3500 candidates—tractable with multiprocessing.

For continuous action spaces, the binary-weighted decomposition scales linearly: d dimensions with b bits each require $d \times b$ independent binary CEGIS rounds. Each round is the same size as a standard 2-action synthesis problem. BipedalWalker’s 12 bits (4×3) are optimized via coordinate descent in 15 passes (3 CEGAR rounds $\times$ 5 iterations), with $\sim 3,000$ features evaluated per bit—well within the tractability of the existing pipeline.

15.6 Relationship to Deep RL

The policies synthesized by CSHRL occupy a fundamentally different point in the design space than those produced by deep RL (DQN, PPO, SAC). We do not claim superior performance—a well-tuned PPO achieves comparable or better reward on most of these benchmarks. What CSHRL provides is a qualitatively different *kind* of artifact.

Information content. The most striking result is the compression ratio. All thirteen environments are solved with 1–12 Boolean predicates, encoding the complete decision structure in approximately 18–100 bits. The CartPole policy is a single linear inequality: $2\theta + \dot{\theta} < 0$. The LunarLander policy is three depth-1 predicates. Even BipedalWalker—a continuous-action locomotion task with 24-dimensional state—requires only 12 Boolean predicates. By contrast, a minimal DQN for CartPole requires $\sim 10,000$ floating-point parameters; PPO policies for BipedalWalker use $\sim 50,000$ – $500,000$ parameters. This is not merely an engineering convenience—it is a scientific finding about the environments themselves: *the decision structure of these RL benchmarks is fundamentally low-dimensional*, and deep RL’s parameter overhead is spent on representation, not on decision complexity.

Interpretability. A human can read $2\theta + \dot{\theta} < 0 \Rightarrow \text{Left}$ and immediately understand *why*: it is a phase-space hyperplane separating the “falling left” from “falling right” regimes. Failure modes, edge cases, and domain transfer properties are visible by inspection. This interpretability is not an add-on or a post-hoc explanation—it is the policy itself. No neural network policy, regardless of architecture or training, provides this transparency.

Formal verifiability (where applicable). The synthesized predicates are PredProg terms in a Boolean DSL amenable to Agda verification. The companion paper [2] demonstrates end-to-end `refl` proofs for CartPole, MountainCar, and Acrobot policies under Agda’s embedded dynamics. For the simulation-based pipeline, the main-engine predicate of LunarLander is verified (`main-cert = refl`); the side-engine predicates exhibit a sim-to-sim gap between the Agda Euler model and the Box2D engine. In general, Synthex-discovered policies are *verifiable in principle* (they are PredProg terms) but *not automatically verified* (the Agda

dynamics model must exist and match the real environment). This is an honest limitation of the hybrid approach.

What deep RL does better. Deep RL handles high-dimensional observations (pixels) and complex multi-modal reward landscapes where Boolean predicates over hand-extractable features are insufficient. Our approach requires a compact state representation (RAM bytes, physics variables). BipedalWalker, Swimmer, and Hopper (§11) demonstrate that continuous action spaces are no longer a fundamental barrier: binary-weighted decomposition maps continuous torque control to 6–12 independent binary predicates, achieving near-solved performance (296.8/ep, 351.6/ep, 1098.2/ep). The remaining genuine limitation is the reliance on extractable state features—pixel observations would require a separate feature extraction layer. The two paradigms are complementary: deep RL scales to raw-pixel domains CSHRL cannot yet reach, while CSHRL provides interpretability and formal verification guarantees deep RL cannot offer.

16 Conclusion

We have presented Synthex, a CSHRL-inspired synthesis engine that discovers compact Boolean-predicate policies for standard RL environments. The key contributions:

1. **Honest hybrid architecture.** Synthex uses CSHRL’s conceptual framework—rankings, coinductive consistency, predicate programs—but replaces formal stream comparisons with heuristic episode-reward scoring. This trades formal guarantees for empirical scalability to environments Agda cannot model natively.
2. **A landscape of synthesis strategies.** Four methods (flat-partition CEGAR, coordinate descent, partition search, binary-weighted decomposition) handle different problem structures. No single method dominates; the choice is environment-dependent (Table 1).
3. **Compactness as a scientific finding.** All thirteen environments are solved with 1–12 Boolean predicates and zero neural network parameters. This is not an engineering constraint but an empirical discovery: standard RL benchmarks have fundamentally low-dimensional decision structure.
4. **Continuous actions via binary decomposition.** BipedalWalker-v3 is controlled by 12 Boolean predicates (3 bits $\times$ 4 joints), achieving 296.8/ep mean reward. Swimmer-v5 (MuJoCo) is controlled by 6 predicates (3 bits $\times$ 2 joints), achieving 351.6/ep with 100% survival. Hopper-v5 (MuJoCo) is controlled by 9 predicates (3 bits $\times$ 3 joints), achieving 1098.2/ep with 97% survival. The decomposition generalizes across physics engines (Box2D, MuJoCo) and locomotion strategies.
5. **Cross-validation with formal synthesis.** CliffWalking and FrozenLake 8 $\times$ 8 are independently solved by both Synthex and the pure-Agda pipeline [2], converging on the same policy structure. This provides evidence that the heuristic search is finding genuinely good policies, even without formal guarantees.

The main limitation is clear: Synthex does not verify that discovered rankings are true coinductive homomorphisms. The policies work empirically, but their correctness is validated by deployment, not by proof. Where Agda verification succeeds (CartPole, MountainCar, Acrobot, LunarLander’s main-engine predicate), it provides an additional layer of confidence. Closing the gap between Synthex’s reach and the formal pipeline’s guarantees remains open.

References

- [1] Corral-Corral, R. (2026). Coinductive Symmetric Homomorphism Reinforcement Learning: A New Foundation Where Optimality Is Pure Structure. *Literate Agda manuscript*.

- [2] Corral-Corral, R. (2026). Verified Predicate Synthesis for Coinductive Ranking: From CEGIS to Compositional Algebra. *Literate Agda manuscript*.